

Computer Graphics in Automotive Design

1 Introduction

This project investigates how three-dimensional images are effectively rendered in two dimensions. This operation is crucial in automotive design, as it allows engineers to experiment with images on a computer screen instead of using a three-dimensional model of the automobile.

The data for this project was derived from measurements made on a 1983 Toyota Corolla, with all coordinates measured in feet. The origin is placed at the center of the car. The data points are: $(-6.5, -2, -2.5)$, $(-6.5, -2, 2.5)$, $(-6.5, .5, 2.5)$, $(-6.5, .5, -2.5)$, $(89-2.5, 5, -2.5)$, $(-2.5, .5, 2.5)$, $(-.75, 2, -2.5)$, $(-.75, 2, 2.5)$, $(3.25, 2, -2.5)$, $(3.25, 2, 2.5)$, $(4.5, 5, -2.5)$, $(4.5, .5, 2.5)$, $(6.5, 5, 2.5)$, $(6.5, .5, 2.5)$, $(6.5, -2, 2.5)$, $(6.5, -2, -2.5)$.

These data points are collected in a data matrix D , where each column contains the x , y , and z coordinates of a particular data point. A fourth row containing all ones is attached to the matrix since homogeneous coordinates are being used:

$$D = \begin{bmatrix} -6.5 & -6.5 & -6.5 & -6.5 & -2.5 & -2.5 & -0.75 & -0.75 & 3.25 & 3.25 & 4.5 & 4.5 & 6.5 & 6.5 & 6.5 & 6.5 \\ -2 & -2 & 0.5 & 0.5 & 0.5 & 0.5 & 2 & 2 & 2 & 2 & 0.5 & 0.5 & 0.5 & 0.5 & -2 & -2 \\ -2.5 & 2.5 & 2.5 & -2.5 & -2.5 & 2.5 & -2.5 & 2.5 & -2.5 & 2.5 & -2.5 & 2.5 & -2.5 & 2.5 & 2.5 & -2.5 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

In addition to the data points, the connections between these points must be known. An adjacency matrix is used to record the connections between points. This matrix consists only of 0s and 1s, where the (i, j) -entry is 1 if and only if points i and j are connected. The adjacency matrix for the Toyota Corolla data is:

$$C = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix}.$$

2 Perspective Projection

A common way to render a three-dimensional image in two dimensions is a perspective projection, as discussed in Section 2.7. To perform a perspective projection, a center of projection (b, c, d) and a viewing plane are identified. The center of projection represents the viewer's eye position, while the viewing plane is the plane onto which the image is projected.

In this case study, it is assumed that the viewing plane is the xy -plane, and any choice of center of projection (b, c, d) is allowed.

3 Questions

1. (10 points) Compute the perspective projection of the Toyota data in matrix D using $(b, c, d) = (-5, 10, 10)$ as the center of projection. Determine the projection matrix and include a figure generated using Python.
2. (10 points) Repeat the perspective projection with $(b, c, d) = (0, 10, 25)$ as the center of projection. Include the computed projection matrix and the corresponding figure.
3. (10 points) Rotate the Toyota by 30° about the y -axis before performing the perspective projection with $(b, c, d) = (0, 10, 25)$. Compute the rotation matrix, include the transformed figure, and compare it with the previous figure by plotting two figures side-by-side.
4. (10 points) Rotate the Toyota by 45° about the z -axis before performing the perspective projection with $(b, c, d) = (0, 10, 25)$. Compute the rotation matrix, include the transformed figure, and compare it with the projection in part 2 by plotting two figures side-by-side.
5. (10 points) Apply a zoom factor of 150% to the Toyota data before performing the perspective projection with the center of projection $(b, c, d) = (0, 10, 25)$. Compute the zoom matrix, include the transformed figure, and compare it with the original projection from part 2 by plotting the two side-by-side.

4 Instructions for Submission

1. Only the following Python libraries are allowed: `math`, `numpy`, `matplotlib`, and `sympy`.
2. Answer all questions in order, ensuring that all Python-generated figures are properly labeled.
3. Prepare a report in Word, Google Docs, or $\text{\LaTeX}$, then convert it into a PDF and submit it via Canvas.
4. (50 points) Submit your Python code as a single `.py` file. The program will be tested for grading.